Extending the Factorial

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Introduction

One of the most interesting aspects of mathematics is the extension of certain concepts into larger sets of numbers. For instance, we are all initially taught to think of exponentiation as repeated multiplication. For example,

$$2^3 = 2 \times 2 \times 2.$$

This definition of exponentiation as the repeated multiplication of the base only makes sense for whole-number exponents. However, mathematicians then figured out ways of making sense of rational, irrational, imaginary, and even matrix exponents. I believe this way of thinking leads to the discovery of the most beautiful results in all of mathematics. One such beautiful result is Euler's identity:

$$e^{i\pi} = -1.$$

Mathematicians thought of another concept that could produce interesting results when extended into real numbers from natural numbers: the factorial of a natural number n, denoted by $n!$.
The factorial of n is defined in natural numbers as follows:

$$n! \coloneqq n(n-1)(n-2)\cdots 3 \cdot 2 \cdot 1 = \prod_{i=1}^{n} i$$

so,

$$n! = n(n-1)!.$$

This recursive formula allows us to calculate the factorials of natural numbers. Plugging in $n = 1$, we get:

$$1! = 1 \cdot (1-1)!,$$

which means that $0! = 1$. In general, if the factorial of a natural number is already known, the next or previous natural number's factorial can easily be calculated using the recursive formula.
Factorials show up in the series expansions of various functions, combinatorics, probability, computer science, and in many more fields of mathematics and science. One of the most important and popular applications of the factorial is in calculating the number of arrangements of multiple items: n items can be arranged in $n!$ ways. Another use of the factorial is in the estimation of the number e, using the formula

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots \approx 2.718281828459.$$

After being told perpetually that the definition of $n!$ only holds for natural numbers in school, I was blown away when I first found out that it was possible to extend it to real numbers. Being my first recognition of mathematical beauty, this was an especially special discovery of mine, because it greatly increased my appreciation and passion for mathematics. I started to improve my understanding of calculus to fully comprehend this concept, growing a larger and larger passion for mathematics along the way.

Deriving the Generalized Definition of the Factorial

When the values of $n!$ are plotted on a graph, discrete points that are not connected by a curve are produced. Generalizing the factorial corresponds to defining a function, which I will denote by $\Pi(x)$, such that it connects the dots on the graph with a curve. The factorial function should have the following properties:

- $\Pi(x) = x!$ for $x \in \mathbb{N}$,
- $\Pi(x) = x\Pi(x-1)$,
- the function should take all real values of x (except for possible values where the function is undefined).

In order to derive $\Pi(x)$, we start by considering the following parameterized integral function:

$$I(s) = \int_0^\infty e^{-st}\,dt.$$

The next step is to differentiate this function repeatedly and look for a pattern. We will make use of the Leibniz integral rule to differentiate under the integral sign (also known as Feynman's integration trick):

$$\begin{aligned}
\frac{dI(s)}{ds} &= \int_0^\infty \frac{\partial}{\partial s}\left(e^{-st}\right)\,dt = \int_0^\infty -te^{-st}\,dt,\\
\frac{d^2I(s)}{ds^2} &= \int_0^\infty \frac{\partial}{\partial s}\left(-te^{-st}\right)\,dt = \int_0^\infty t^2e^{-st}\,dt,\\
\frac{d^3I(s)}{ds^3} &= \int_0^\infty \frac{\partial}{\partial s}\left(t^2e^{-st}\right)\,dt = \int_0^\infty -t^3e^{-st}\,dt,
\end{aligned}$$

so, in general:

$$\frac{d^nI(s)}{ds^n} = \int_0^\infty (-1)^n t^n e^{-st}\,dt = (-1)^n\mathcal{L}\{t^n\}.$$

The expression for the nth derivative of $I(s)$ is equivalent to $(-1)^n\mathcal{L}\{t^n\}$, where $\mathcal{L}\{t^n\}$ is the Laplace transform of t^n. In order to complete our derivation, we must first find an expression for $\mathcal{L}\{t^n\}$. We will make use of

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a},$$

and the series expansions of the functions $\frac{1}{1-x}$ and e^{ax}.
We have:

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a} = \mathcal{L}\left\{\sum_{n=0}^{\infty}\frac{a^nt^n}{n!}\right\},$$

and,

$$\frac{1}{s-a} = \frac{1}{s} \cdot \frac{1}{1-a/s} = \frac{1}{s} \sum_{n=0}^{\infty} \left(\frac{a}{s}\right)^n = \sum_{n=0}^{\infty} \frac{a^n}{s^{n+1}}.$$

Assuming that the Laplace transform and the summation are interchangeable:

$$\sum_{n=0}^{\infty} \frac{a^n}{n!} \mathcal{L}\{t^n\} = \sum_{n=0}^{\infty} \frac{a^n}{s^{n+1}}.$$

Since both sides are sums with the same indices, we can conclude that:

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}.$$

In the beginning of the derivation, we found out that the function $I(s)$'s nth derivative is $(-1)^n \mathcal{L}\{t^n\}$, which implies:

$$(-1)^n \mathcal{L}\{t^n\} = \frac{(-1)^n n!}{s^{n+1}} = (-1)^n \int_0^{\infty} t^n e^{-st}\, dt.$$

Hence,

$$\frac{n!}{s^{n+1}} = \int_0^{\infty} t^n e^{-st}\, dt.$$

Now make the substitution $u = st$, so that $du = s\, dt$. Then,

$$\frac{n!}{s^{n+1}} = \frac{1}{s^{n+1}} \int_0^{\infty} u^n e^{-u}\, du,$$

which implies:

$$n! = \int_0^{\infty} t^n e^{-t}\, dt.$$

We have finally derived an expression that represents the factorial as an integral. Therefore, we can now conclude that our generalized factorial function,

$$\Pi(x) := \int_0^{\infty} t^x e^{-t}\, dt, \quad x > -1$$

can be used to calculate the factorial of any real number. $\Pi(x)$ also satisfies the conditions listed earlier. I will not go through the process of proving these properties, because the proofs do not add much to the final conclusion.

Note: This derivation can be made more rigorous by justifying the interchange of limits and providing the conditions under which the infinite sums converge.
A slightly altered version of $\Pi(x)$ that is more commonly used in the mathematical community, known as the Gamma function, $\Gamma(x)$, is defined as follows:

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t}\, dt.$$

So $\Gamma(x) = \Pi(x-1)$, and $\Gamma(n) = (n-1)!$ for whole numbers. The Gamma function is the standard extension of the factorial used in modern mathematics.

Alternative Representations of the Gamma Function

Another peculiar thing about the Gamma function is that it can be represented in many different forms. Let us take a look at different representations of the Gamma function.

1. Gauss Representation

Using the limit definition of the exponential function, the integral representation of the Gamma function can be transformed into the following:

$$\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}\,dt = \int_0^\infty t^{x-1} \lim_{n\to\infty}\left(1-\frac{t}{n}\right)^n dt.$$

Assuming that the limit and integration are interchangeable, we get:

$$\Gamma(x) = \lim_{n\to\infty}\int_0^\infty t^{x-1}\left(1-\frac{t}{n}\right)^n dt.$$

After doing $n+1$ iterations of integration by parts and evaluating the anti derivative at the limits of integration (which I will not demonstrate to avoid unnecessarily prolonging this derivation), we get:

$$\Gamma(x) = \lim_{n\to\infty}\frac{n^x}{x}\cdot\frac{1\cdot 2\cdots(n-1)\cdot n}{(x+1)(x+2)\cdots(x+n-1)(x+n)}.$$

Expressing the product in capital pi notation, the following final form is obtained:

$$\Gamma(x) = \frac{1}{x}\lim_{n\to\infty} n^x \prod_{k=1}^{n}\frac{k}{x+k}.$$

2. Euler Representation

The number n can be represented in the following way:

$$n = \frac{2}{1}\cdot\frac{3}{2}\cdot\frac{4}{3}\cdots\frac{n}{n-1} = \prod_{k=1}^{n-1}\frac{k+1}{k}.$$

Multiplying the product with the term at n and its reciprocal:

$$n = \prod_{k=1}^{n-1}\frac{k+1}{k}\cdot\frac{n+1}{n}\cdot\frac{n}{n+1} = \frac{n}{n+1}\prod_{k=1}^{n}\frac{k+1}{k}.$$

Plugging this representation of n into the Gauss representation gives the Euler representation:

$$\Gamma(x) = \frac{1}{x}\lim_{n\to\infty}\left(1-\frac{1}{n+1}\right)^x\prod_{k=1}^{n}\left(1+\frac{1}{k}\right)^x\prod_{k=1}^{n}\frac{k}{x+k}.$$

Hence,

$$\Gamma(x) = \frac{1}{x}\prod_{k=1}^{\infty}\frac{\left(1+\frac{1}{k}\right)^x}{1+\frac{x}{k}}.$$

3. Weierstrass Representation

The final and most interesting representation of the Gamma function is due to Weierstrass. We will start with the Gauss representation and write n^x as $\exp(x\ln n)$:

$$\Gamma(x) = \frac{1}{x}\lim_{n\to\infty}\exp(x\ln n)\prod_{k=1}^{n}\frac{k}{x+k}.$$

Now add and subtract the harmonic number $H_n = \sum_{k=1}^{n} \frac{1}{k}$ in the exponent:

$$\Gamma(x) = \frac{1}{x} \lim_{n\to\infty} \exp(xH_n - xH_n + x\ln n) \prod_{k=1}^{n} \frac{k}{x+k}$$

Further algebraic manipulation yields:

$$\Gamma(x) = \frac{1}{x} \lim_{n\to\infty} \prod_{k=1}^{n} e^{x/k} \cdot \exp\big(x(\ln n - H_n)\big) \cdot \prod_{k=1}^{n} \frac{k}{x+k}.$$

The limiting difference between the harmonic series and the natural logarithm is equal to what is known as the Euler–Mascheroni constant.

$$\gamma := \lim_{n\to\infty} (H_n - \ln n).$$

With this information, we can conclude that:

$$\Gamma(x) = \frac{e^{-\gamma x}}{x} \prod_{k=1}^{\infty} e^{x/k} \left(1 + \frac{x}{k}\right)^{-1}.$$

These different representations are going to be useful when deriving relationships involving the Gamma function, one of which is Euler's reflection formula.

Euler's Reflection Formula

Perhaps the most important and useful functional equation of the Gamma function is Euler's reflection formula, which I will derive now using the Euler representation of the Gamma function along with the infinite product representation of $\sin \pi x$.
We will begin by considering $\Gamma(x)\Gamma(1-x)$, which is equivalent to $(-x)\Gamma(-x)\Gamma(x)$. Using the Euler product:

$$-x\Gamma(-x)\Gamma(x) = \frac{1}{x} \prod_{k=1}^{\infty} \frac{1}{1 - x^2/k^2}.$$

This product is very similar to the infinite product representation of $\sin \pi x$, which is:

$$\sin \pi x = \pi x \prod_{k=1}^{\infty} \left(1 - \frac{x^2}{k^2}\right).$$

Taking reciprocals of both sides gives:

$$\frac{1}{\sin \pi x} = \frac{1}{\pi x} \prod_{k=1}^{\infty} \frac{1}{1 - x^2/k^2},$$

and multiplying both sides by π gives:

$$\frac{\pi}{\sin \pi x} = \frac{1}{x} \prod_{k=1}^{\infty} \frac{1}{1 - x^2/k^2}.$$

In conclusion,

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x},$$

or, equivalently:

$$\Gamma(x)\Gamma(1-x) = \pi \csc(\pi x).$$

We can use this result to calculate the value of $\left(-\frac{1}{2}\right)!$. Plugging in $x = \frac{1}{2}$ into Euler's reflection formula gives:

$$\Gamma\left(\frac{1}{2}\right)\Gamma\left(1-\frac{1}{2}\right) = \pi \csc\left(\frac{\pi}{2}\right) = \pi,$$

so,

$$\Gamma\left(\frac{1}{2}\right)^2 = \pi,$$

and therefore:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

This surprising result enables us to determine other values of the Gamma function using its recursive definition. For example:

$$\Gamma\left(\frac{3}{2}\right) = \Gamma\left(\frac{1}{2}+1\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right),$$

so,

$$\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2},$$

and, similarly:

$$\Gamma\left(\frac{5}{2}\right) = \Gamma\left(\frac{3}{2}+1\right) = \frac{3}{2}\Gamma\left(\frac{3}{2}\right) = \frac{3\sqrt{\pi}}{4}.$$

Gamma Function's Relation to Other Functions

There are interesting equations that relate the Gamma function to other special functions. These relationships are often quite surprising, because it is usually not immediately obvious that the Gamma function is related to other famous functions. The functions I would like to explore now are the following:

1. Riemann Zeta Function

We will start with the following integral:

$$\int_0^\infty \frac{t^{x-1}}{e^t - 1}\, dt.$$

The first step is to multiply the integrand's numerator and denominator by e^{-t}:

$$\int_0^\infty \frac{t^{x-1}}{e^t - 1}\, dt = \int_0^\infty \frac{t^{x-1}e^{-t}}{1-e^{-t}}\, dt.$$

Using

$$\frac{1}{1-r} = \sum_{k=0}^{\infty} r^k,$$

we can rewrite $\frac{1}{1-e^{-t}}$ as $\sum_{k=0}^{\infty} e^{-kt}$. Plugging this into the integral gives:

$$\int_0^\infty \frac{t^{x-1}e^{-t}}{1-e^{-t}}\, dt = \int_0^\infty \sum_{k=0}^{\infty} e^{-kt}t^{x-1}e^{-t}\, dt.$$

Once again, assuming that the order of summation and integration are interchangeable, we can rewrite the integral as:

$$\sum_{k=0}^{\infty} \int_0^{\infty} t^{x-1} e^{-(k+1)t}\, dt.$$

Now, let $u = (k+1)t$, so that $du = (k+1)dt$. Then:

$$\sum_{k=0}^{\infty} \frac{1}{(k+1)^x} \int_0^{\infty} u^{x-1} e^{-u}\, du.$$

Since

$$\sum_{k=0}^{\infty} \frac{1}{(k+1)^x} = \sum_{k=1}^{\infty} \frac{1}{k^x} = \zeta(x),$$

and,

$$\Gamma(x) = \int_0^{\infty} u^{x-1} e^{-u}\, du,$$

we obtain:

$$\int_0^{\infty} \frac{t^{x-1}}{e^t - 1}\, dt = \zeta(x)\Gamma(x),$$

where $\zeta(x)$ is the Riemann zeta function. This relationship provides another alternative representation of the Gamma function in terms of the Riemann zeta function:

$$\Gamma(x) = \frac{1}{\zeta(x)} \int_0^{\infty} \frac{t^{x-1}}{e^t - 1}\, dt.$$

Not only does this relationship provide a new way of expressing the Gamma function, but it also enables us to find a general solution of the integral for different values of x. For instance, it is well known that $\zeta(2) = \frac{\pi^2}{6}$ and $\Gamma(2) = 1$. Using this we get

$$1 = \frac{6}{\pi^2} \int_0^{\infty} \frac{t}{e^t - 1}\, dt,$$

so,

$$\int_0^{\infty} \frac{t}{e^t - 1}\, dt = \frac{\pi^2}{6}.$$

2. Dirichlet Eta Function

A similar process can be carried out to obtain a similar equation that links the Gamma function with the Dirichlet eta function. The result is the following:

$$\int_0^{\infty} \frac{t^{x-1}}{e^t + 1}\, dt = \eta(x)\Gamma(x),$$

where the Dirichlet eta function is defined as:

$$\eta(x) := \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k^x}.$$

Using the fact that

$$\eta(x) = (1 - 2^{1-x})\zeta(x),$$

we can rewrite this equation as:

$$\int_0^\infty \frac{t^{x-1}}{e^t+1}\,dt = (1-2^{1-x})\zeta(x)\Gamma(x).$$

For example, letting $x = 2$ yields:

$$\int_0^\infty \frac{t}{e^t+1}\,dt = \eta(2)\Gamma(2) = (1-2^{1-2})\zeta(2)\Gamma(2),$$

so,

$$\int_0^\infty \frac{t}{e^t+1}\,dt = \frac{\pi^2}{12}.$$

3. Digamma Function

The digamma function is defined as the derivative of the natural logarithm of $\Gamma(x)$.

$$\psi(x) := \frac{d}{dx}\ln\Gamma(x).$$

Using the Weierstrass definition of the Gamma function, we can express the reciprocal of the Gamma function as follows:

$$\frac{1}{\Gamma(x)} = xe^{\gamma x}\prod_{k=1}^{\infty}\left(\frac{k+x}{k}\right)e^{-x/k}.$$

Taking the natural logarithm of both sides gives:

$$-\ln\Gamma(x) = \ln x + \gamma x + \sum_{k=1}^{\infty}\ln\left(\frac{k+x}{k}\right) - \frac{x}{k},$$

so,

$$\ln\Gamma(x) = -\ln x - \gamma x + \sum_{k=1}^{\infty}\left(\frac{x}{k} - \ln\left(\frac{k+x}{k}\right)\right).$$

Differentiating both sides gives the following expression for the digamma function:

$$\psi(x) = -\frac{1}{x} - \gamma + \sum_{k=1}^{\infty}\left(\frac{1}{k} - \frac{1}{k+x}\right).$$

The derivative of the Gamma function shows up frequently in different branches of mathematics. In fact, it is commonly represented in terms of the digamma function to make its evaluation easier at particular values of x. This representation is as follows:

$$\Gamma'(x) = \Gamma(x)\psi(x).$$

Calculating the values of $\psi(x)$ at integer values of x is an interesting process, one that involves harmonic series, which we have previously discussed when deriving the Weierstrass representation of $\Gamma(x)$.

To find a general expression for $\psi(n)$, where n is an integer, we start by splitting the sum in the definition of $\psi(x)$ as follows:

$$\psi(n) = -\frac{1}{n} - \gamma + \sum_{k=1}^{n}\left(\frac{1}{k} - \frac{1}{k+n}\right) + \sum_{k=n+1}^{\infty}\left(\frac{1}{k} - \frac{1}{k+n}\right).$$

It becomes apparent that the sums are telescoping once we look at the partial sums. The cancellation leaves the harmonic sum, producing

$$\psi(n) = -\frac{1}{n} - \gamma + \sum_{k=1}^{n} \frac{1}{k}.$$

Since the final term in the sum is $1/n$, it cancels with the $-1/n$ term, giving the final equation:

$$\psi(n) = -\gamma + \sum_{k=1}^{n-1} \frac{1}{k} = -\gamma + H_{n-1}.$$

We can therefore express the derivative of the Gamma function at integer values as:

$$\Gamma'(n) = (-\gamma + H_{n-1})(n-1)!.$$

For example:

$$\Gamma'(2) = (-\gamma + H_1)(2-1)! = 1 - \gamma,$$

and,

$$\Gamma'(3) = (-\gamma + H_2)(3-1)! = 3 - 2\gamma.$$

4. Beta Function (Euler Integral of the First Kind)

The Beta function is a multivariable function defined in terms of the Gamma function. To derive it, we start with the following product of two Gamma functions:

$$\Gamma(x)\Gamma(y) = \int_0^\infty t^{x-1}e^{-t}\,dt \int_0^\infty r^{y-1}e^{-r}\,dr = \iint_{(0,\infty)^2} t^{x-1}r^{y-1}e^{-(t+r)}\,dr\,dt.$$

Introduce the substitution:

$$r = uv, \qquad t = u(1-v).$$

To determine the new limits of integration, notice that r and t are both greater than 0. This means that u and v too must be greater than 0. However, v must also be less than 1, because otherwise t would be less than 0. So, $u > 0$ and $0 < v < 1$. The determinant of the Jacobian matrix is:

$$\begin{vmatrix} \dfrac{\partial r}{\partial v} & \dfrac{\partial r}{\partial u} \\ \dfrac{\partial t}{\partial v} & \dfrac{\partial t}{\partial u} \end{vmatrix} = \begin{vmatrix} u & v \\ -u & 1-v \end{vmatrix} = u.$$

Therefore,

$$\Gamma(x)\Gamma(y) = \int_0^\infty \int_0^1 u^{x+y-1}e^{-u}v^{y-1}(1-v)^{x-1}\,du\,dv.$$

Separating the integrals gives:

$$\Gamma(x)\Gamma(y) = \int_0^\infty u^{x+y-1}e^{-u}\,du \int_0^1 v^{y-1}(1-v)^{x-1}\,dv.$$

Using the integral definition of the Gamma function, we deduce that:

$$\Gamma(x)\Gamma(y) = \Gamma(x+y)\int_0^1 v^{y-1}(1-v)^{x-1}\,dv,$$

where,

$$\int_0^1 v^{y-1}(1-v)^{x-1}\,dv$$

is defined to be the Beta function, $B(x,y)$. We can therefore express the Beta function in terms of the Gamma function as:

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

The Beta function also has an equivalent trigonometric form, which is extremely useful when evaluating trigonometric integrals. To derive it we use the substitution $v = \sin^2\theta$, $dv = 2\sin\theta\cos\theta\,d\theta$:

$$\int_0^1 v^{y-1}(1-v)^{x-1}\,dv = \int_0^{\pi/2} \sin^{2y-2}\theta\cos^{2x-2}\theta\cdot 2\sin\theta\cos\theta\,d\theta,$$

so,

$$B(x,y) = 2\int_0^{\pi/2} \sin^{2y-1}\theta\cos^{2x-1}\theta\,d\theta.$$

Notice that the places of x and y can be switched without changing the result, due to the Beta function's symmetry.
To demonstrate the practicality of this form of the Beta function, I would like to use it to solve an integral which requires a relatively large amount of tedious work to be solved using standard methods of integration. Consider:

$$\int_0^{\pi/2} \sqrt{\tan x}\,dx = \int_0^{\pi/2} \sin^{1/2}x\cos^{-1/2}x\,dx.$$

When written in this form, it is apparent that the integral is equal to $\frac{1}{2}B\left(\frac{3}{4},\frac{1}{4}\right)$. Therefore:

$$\int_0^{\pi/2} \sin^{1/2}x\cos^{-1/2}x\,dx = \frac{1}{2}B\left(\frac{3}{4},\frac{1}{4}\right) = \frac{\Gamma(3/4)\Gamma(1/4)}{2\Gamma(1)}.$$

Using Euler's reflection formula:

$$\frac{\Gamma(3/4)\Gamma(1/4)}{2} = \frac{1}{2}\Gamma\left(\frac{3}{4}\right)\Gamma\left(1-\frac{3}{4}\right) = \frac{\pi}{2}\csc\left(\frac{3\pi}{4}\right) = \frac{\pi\sqrt{2}}{2}.$$

Hence,

$$\int_0^{\pi/2} \sqrt{\tan x}\,dx = \frac{\pi\sqrt{2}}{2}.$$

Legendre's Duplication Formula

Another beautiful Gamma function identity can be derived using the Beta function. Consider $B(x,x)$:

$$B(x,x) = \frac{\Gamma(x)\Gamma(x)}{\Gamma(2x)} = \int_0^1 t^{x-1}(1-t)^{1-x}\,dt$$

Introduce the substitution $u = \frac{1+t}{2}$, $\quad dt = \frac{1}{2}du$:

$$\int_0^1 t^{x-1}(1-t)^{x-1}\,dt = \frac{1}{2}\int_{-1}^1 \left(\frac{1+u}{2}\right)^{x-1}\left(\frac{1-u}{2}\right)^{x-1}du = 2\cdot 2^{1-2x}\int_0^1 (1-u^2)^{x-1}\,du$$

Introduce another substitution $v = u^2, \quad dv = 2udu$:

$$2 \cdot 2^{1-2x} \int_0^1 (1-u^2)^{x-1}\, du = 2^{1-2x} \int_0^1 (1-v)^{x-1} v^{-\frac{1}{2}}\, dv = 2^{1-2x} B\left(z, \frac{1}{2}\right)$$

So, we get:

$$\frac{\Gamma(x)\Gamma(x)}{\Gamma(2x)} = 2^{1-2x} B\left(x, \frac{1}{2}\right) = 2^{1-2x} \frac{\Gamma(x)\Gamma(\frac{1}{2})}{\Gamma(x+\frac{1}{2})}$$

Plugging in $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, simplifying, and rearranging:

$$\Gamma(x)\Gamma\left(x + \frac{1}{2}\right) = 2^{1-2x}\sqrt{\pi}\Gamma(2x)$$

This identity is known as Legendre's duplication formula. It is a special case of the Gauss Multiplication Theorem, which I will not go into here. As an example, let's consider the case when $x = \frac{1}{4}$:

$$\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right) = 2^{\frac{1}{2}}\sqrt{\pi}\Gamma\left(\frac{1}{2}\right)$$

Which simplifies to:

$$\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right) = \pi\sqrt{2}$$

Stirling's Approximation

An important approximation of the Gamma function is due to Stirling. We will (heuristically) derive it in the following way:
Let $f(x)$ be a function that asymptotically approximates $x!$. Then, we need:

$$\lim_{x\to\infty} \frac{x!}{f(x)} = 1$$

We can write $x!$ as:

$$x! = \Gamma(x+1) = \int_0^\infty t^x e^{-t}\, dt = \int_0^\infty e^{x\ln t - t}\, dt$$

Introduce the substitution $t = (1+y)x, \quad dt = xdy$:

$$\Gamma(x+1) = \int_{-1}^\infty x\exp(x\ln(1+y) + x\ln x - x - xy)\, dy = x^{x+1}e^{-x}\int_{-1}^\infty \exp(x(\ln(1+y) - y))\, dy$$

Since we are looking to approximate $\Gamma(x+1)$, we will investigate the limiting behavior of the integrand. Let $\phi(y) = \ln(1+y) - y$. Notice that $\phi(y) \leq 0$ with equality only at $y = 0$. Also notice that $\phi(y) \approx -\frac{y^2}{2}$ for small values of y. Therefore, we can approximate $x!$ as:

$$x! \sim x^{x+1}e^{-x}\int_{-\infty}^\infty e^{-\frac{xy^2}{2}}\, dy$$

Notice that the lower bound has been changed from -1 to $-\infty$. This is because the dominant contribution to the integral comes from the neighborhood of $y = 0$. So, extending the lower limit to $-\infty$ does not affect the limiting behavior. The integral that we have obtained is of the form of the Gaussian integral (which we will evaluate in the next section) up to a simple substitution. So:

$$\int_{-\infty}^\infty e^{-\frac{xy^2}{2}}\, dy = \sqrt{\frac{2}{x}}\int_{-\infty}^\infty e^{-y^2}\, dy = \sqrt{\frac{2\pi}{x}}$$

Plugging in:

$$x! \sim x^x e^{-x}\sqrt{2\pi x}$$

Which is precisely Stirling's formula. This is an example of Laplace's method for approximating integrals.

Interesting Problems Involving the Gamma Function

The Gamma function and the other identities that we have derived so far come in handy when solving seemingly unrelated problems. Making use of the Gamma function simplifies many problems and provides beautiful solutions. I would now like to work through a few problems to demonstrate this.

Problem 1

Evaluate the following generalized integral:

$$I = \int_0^\infty \frac{dx}{x^n + 1}.$$

To solve this problem, we will make use of the Leibniz integral rule as we did before. Let

$$I(t) = \int_0^\infty \frac{1}{x^n + 1} e^{-(x^n+1)t}\, dx.$$

Introducing the parameter t in this way is a little counterintuitive, but it makes the calculation easier. Differentiate with respect to t:

$$\frac{dI(t)}{dt} = \int_0^\infty \frac{\partial}{\partial t}\left(\frac{1}{x^n + 1} e^{-(x^n+1)t}\right) dx = -\int_0^\infty e^{-(x^n+1)t}\, dx.$$

Rewrite this as:

$$\frac{dI(t)}{dt} = -e^{-t}\int_0^\infty e^{-tx^n}\, dx.$$

Now use the substitution $u = tx^n$, so that $du = tnx^{n-1}\, dx$, which yields:

$$\frac{dI(t)}{dt} = -\frac{1}{n} t^{-1/n} e^{-t}\int_0^\infty u^{1/n-1} e^{-u}\, du.$$

Since

$$\int_0^\infty u^{1/n-1} e^{-u}\, du = \Gamma\left(\frac{1}{n}\right),$$

we get:

$$\frac{dI(t)}{dt} = -\frac{1}{n} t^{-1/n} e^{-t}\Gamma\left(\frac{1}{n}\right).$$

To get back to the goal integral I, we integrate $\frac{dI(t)}{dt}$ with respect to t from 0 to ∞. This gives:

$$I = \frac{1}{n}\Gamma\left(\frac{1}{n}\right)\int_0^\infty t^{-1/n} e^{-t}\, dt.$$

Since

$$\int_0^\infty t^{-1/n} e^{-t}\, dt = \Gamma\left(1 - \frac{1}{n}\right),$$

we obtain:

$$I = \frac{1}{n}\Gamma\left(\frac{1}{n}\right)\Gamma\left(1-\frac{1}{n}\right).$$

Using Euler's reflection formula, we conclude that:

$$\int_0^\infty \frac{dx}{x^n+1} = \frac{\pi}{n}\csc\left(\frac{\pi}{n}\right).$$

Problem 2

Find the Laplace transform of the natural logarithm of t, namely $\mathcal{L}\{\ln t\}$.
By definition,

$$\mathcal{L}\{\ln t\} = \int_0^\infty \ln t\, e^{-st}\, dt.$$

Use the substitution $u = st$, so that $du = s\, dt$. Then

$$\mathcal{L}\{\ln t\} = \frac{1}{s}\int_0^\infty \ln\left(\frac{u}{s}\right) e^{-u}\, du = \frac{1}{s}\left[\int_0^\infty \ln u\, e^{-u}\, du - \ln s \int_0^\infty e^{-u}\, du\right].$$

The hardest part of finding $\mathcal{L}\{\ln t\}$ is evaluating

$$\int_0^\infty \ln u\, e^{-u}\, du,$$

which cannot be solved using standard methods of integration. To solve this integral, we will first take a look at the derivative of the Gamma function. We once again use differentiation under the integral sign:

$$\Gamma'(x) = \int_0^\infty \frac{\partial}{\partial x}\left(t^{x-1}e^{-t}\right)\, dt = \int_0^\infty t^{x-1}\ln t\, e^{-t}\, dt.$$

It can be seen that

$$\int_0^\infty \ln u\, e^{-u}\, du$$

is equal to the derivative of the Gamma function at $x = 1$. To evaluate $\Gamma'(1)$, we use the formula derived above:

$$\Gamma'(1) = (-\gamma + H_0)(0)! = -\gamma.$$

So,

$$\int_0^\infty \ln u\, e^{-u}\, du = -\gamma.$$

This finally allows us to find the Laplace transform of $\ln t$:

$$\mathcal{L}\{\ln t\} = \frac{1}{s}\left[-\gamma - \ln s \int_0^\infty e^{-u}\, du\right] = -\frac{\gamma + \ln s}{s}.$$

Problem 3

Evaluate the Gaussian integral

$$\int_{-\infty}^\infty e^{-x^2}\, dx,$$

using the Gamma function.
We can use the fact that:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

to calculate the value of the Gaussian integral. Let $x = \sqrt{t}$, so that $dx = \frac{dt}{2\sqrt{t}}$. Then,

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = 2\int_0^{\infty} e^{-x^2}\,dx = 2\int_0^{\infty} \frac{1}{2\sqrt{t}} e^{-t}\,dt = \int_0^{\infty} t^{-1/2} e^{-t}\,dt.$$

Hence,

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

I believe this is a very elegant method of evaluating the Gaussian integral, because it does not rely on polar coordinates and double integrals.

Conclusion

The interesting problem of finding a generalization of the factorial led to finding a function that is an extension of the factorial to real numbers, namely $\Gamma(x)$. We then found four different representations of this function, and used them to derive an interesting result: Euler's reflection formula, which relates a product of gamma functions to a trigonometric function. This alone is a very interesting finding, because the Gamma function and trigonometric functions do not even seem remotely related, yet we managed to connect them somehow.

We then derived two other functions that are closely related to the Gamma function: the Beta function and the Digamma function, both of which proved useful when evaluating certain integrals. Furthermore, we found interesting relations of the Gamma function to the Riemann zeta function and the Dirichlet eta function, which are both extremely important in analytic number theory. Given the fact that the million-dollar problem of the Riemann Hypothesis is all about the Riemann zeta function, I believe the Gamma function plays a central role in the analytic continuation and functional equation of the Riemann zeta function.

We finally put our findings into practice by solving three problems. These problems demonstrated the practicality of the Gamma function and the other results we have arrived at. We combined many other different tools from mathematics along the way, including the Leibniz integral rule, Laplace transforms, and Taylor series.

There are many more applications of the Gamma function, including the gamma distribution from statistics and quantum physics, which I did not even begin to explain. The Gamma function's relations to other functions are not limited to the ones I demonstrated. It is connected to many other functions in many other ways. I also limited the Gamma function's inputs to real numbers; however, it is possible to extend it to complex numbers as well. In short, the Gamma function shows up constantly in mathematics. It was impossible for me to explain everything about it, so I picked my favorites to include.

Despite not including everything I had in mind, I believe I managed to emphasize the importance and beauty of extending a simple concept into a larger set of numbers using the Gamma function as a generalization of the factorial. This is therefore one of my favorite concepts in mathematics.

References

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